Harsha Kanaparthi

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Software engineer with 2.5+ years of industry experience building scalable distributed systems, full-stack applications, and AI-powered features. Shipped systems processing 10K+ events/sec at sub-100ms latency across Python, Go, Node.js, React, AWS, Docker, and Kafka. IEEE-published researcher. M.S. Computer Science, GPA 4.0.

Education

University of North Carolina at Charlotte — Charlotte, NC
M.S. Computer Science (Data Science & InfoSec); GPA: 4.0/4.0 — *Aug. 2024 – May 2026*

Koneru Lakshmaiah University — Vijayawada, India
B.S. Computer Science and Engineering; GPA: 9.47/10.0

Experience

Software Engineer — Jan. 2021 – June 2023
LogicMind

- Designed and shipped scalable REST APIs and real-time gameplay services in Python and Node.js supporting 3K+ concurrent users; owned deployments on AWS (EC2, S3, RDS) with Docker and GitHub Actions CI/CD achieving 99.5% uptime
- Reduced API response latency by 25% through PostgreSQL query optimization, connection pooling, and Redis caching; built event-driven analytics pipelines processing 100K+ gameplay events daily
- Implemented AI-driven matchmaking algorithm improving average session duration by 14%; collaborated cross-functionally in agile sprints to design, test, and ship production features

Graduate Teaching Assistant — Aug. 2025 – May 2026
University of North Carolina at Charlotte — *Charlotte, NC*

- Supported 110+ students in Survey of Programming Languages (Go, Haskell, Swift) and Software Systems Design & Implementation; student A-grade rate improved from 70% to 90% through structured mentoring and feedback
- Designed assignments with 48-hour grading turnaround; implemented lab verification protocols that significantly reduced proxy attendance incidents

AI Intern — June 2023 – Dec. 2023
Lineysha & Thevan Software Company — *Andhra Pradesh, India*

- Built and deployed production ML system for missing child identification using VGG-Face CNN achieving 99.4% facial recognition accuracy across 10K+ images with real-time inference and MySQL backend
- Architected Flask REST API with similarity search across 10K+ face embeddings at <500ms latency; reduced inference time 40% via batch processing and database indexing
- Developed full-stack React frontend with concurrent request handling and role-based access control for students, faculty, and admins; achieved sub-2s end-to-end response

Projects

Stream – Real-Time E-Commerce Analytics | *Kafka, Spark, TimescaleDB, FastAPI, React, Docker* | GitHub — Mar. 2026

- Architected distributed event streaming platform with Apache Kafka (3 brokers) and Spark Structured Streaming processing 10K+ events/sec at <100ms end-to-end latency with exactly-once delivery semantics
- Built automated anomaly detection triggering alerts on statistical deviations across revenue, session, and conversion metrics; deployed via Docker Compose orchestrating 6 services
- Exposed aggregated metrics via FastAPI with WebSocket push to React dashboard; validated throughput under simulated 3x peak load

Biometric Attendance System | *Swift, SwiftUI, Python, Flask, dlib, PostgreSQL, iBeacon* | — Dec. 2025 – Present

- Designed and shipped multi-layer verification combining facial recognition (99%+ accuracy, dlib ResNet-34 CNN), iBeacon Bluetooth proximity (±2m), and liveness detection; replaced QR system after identifying credential-sharing vulnerability
- Built native iOS app (Swift/SwiftUI/AVFoundation/CoreLocation) and Flask backend processing 128-dim face encodings; reduced attendance marking from 60s to under 5s at sub-2s response

SPL – Arithmetic Expression Parser | *Node.js, Express.js, D3.js* | Live Demo | GitHub — Oct. 2025

- Built recursive descent parser with full PEMDAS precedence, generating interactive Parse Tree and AST visualizations via D3.js with expandable nodes, zoom, and pan; deployed live on Vercel
- Designed modular architecture decoupling lexer, parser, and renderer; extended original Java grammar to support subtraction and division without modifying core logic

Technical Skills

Languages: Python, JavaScript/TypeScript, Go, Java, Swift, C++, SQL

Backend & Frameworks: FastAPI, Flask, Express.js, Node.js, Gin, Django, React, SwiftUI, D3.js, REST, WebSockets, gRPC

Databases: PostgreSQL, MySQL, MongoDB, Redis, TimescaleDB

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Docker Compose, GitHub Actions CI/CD, Vercel, Railway | Kafka, Spark Structured Streaming

Other: TensorFlow, PyTorch, OpenCV | iOS (AVFoundation, CoreLocation, BLE/iBeacon) | IEEE ICICV 2024, ICCES 2023 | Competitive Coding Club President